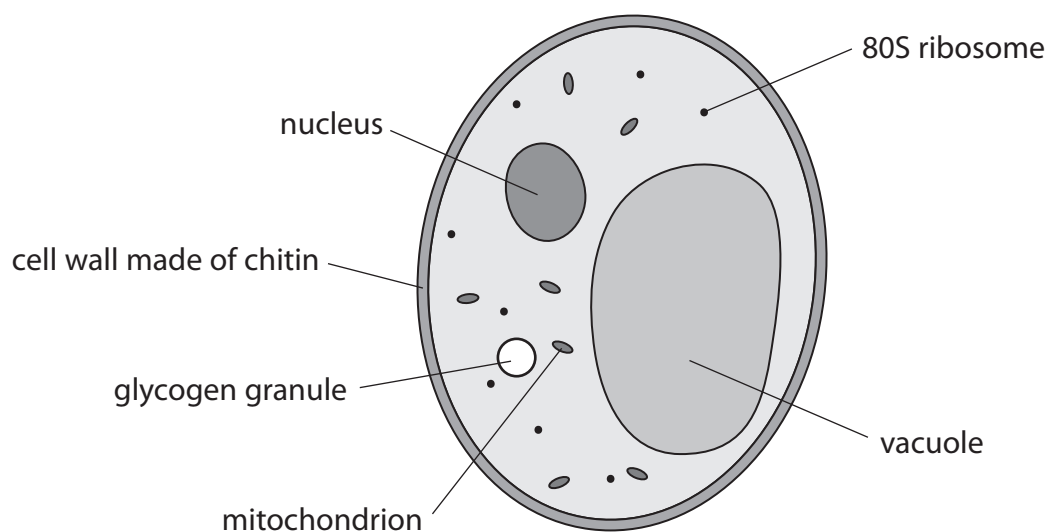


8 Some fungi infect plants and affect the expression of the plant genes.

(a) A student drew and labelled a fungal cell.



Compare and contrast the structure of this fungal cell and a plant cell.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA





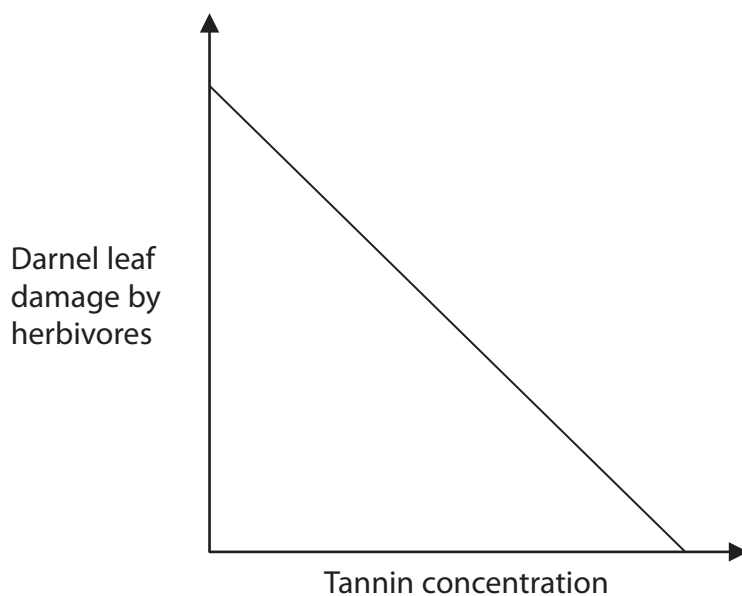
- *(ii) The fungus and the plant both benefit from this relationship.

The fungus absorbs nutrients from the plant.

The expression of certain genes in the plant cells is increased when infected with the fungus.

These genes are involved in the synthesis of tannin and flavonoids.

The graph shows the relationship between tannin concentration and the degree of dandel leaf damage by herbivores (grazing animals).



The table shows the antimicrobial effect of different concentrations of flavonoids on cultures of the bacterium *Pseudomonas aeruginosa*. This bacterium causes disease in plants.

Flavonoid concentration / $\mu\text{g cm}^{-3}$	Diameter of inhibition zone / mm
500	7.0
666	8.0
1000	9.0



The fungus increases the expression of the genes involved in the synthesis of tannin and flavonoids.

Discuss the advantages and disadvantages of this for the fungus and for the infected plants.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



P 6 5 8 1 1 A 0 2 5 2 8

(iii) The bacterium *Pseudomonas aeruginosa* causes lung infections in humans.

Describe how a drug containing flavonoids could be tested in a stage II drug trial.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

